

first token identifies the user as an admin, but is immediately filtered down to a standard user token. Under this standard user token, Vista launches the Explorer.exe process, or the shell inside which the user will work. Since Explorer is now running with standard privileges, and since any application launched from Explorer will inherit those standard privileges, all applications and tasks run by the administrator are now run as standard users. A user who is a member of the Administrators group can now log in, browse the Web, and read his mail while using a standard user access token. When the administrator needs to perform a task that requires the administrator access token, Windows Vista automatically prompts the user for approval.

Windows Vista protects processes by marking them with integrity levels. Integrity levels are measurements of trust. A “high” integrity application can modify system data, while a “low” integrity application might be a Web browser. Vista prevents applications with lower integrity levels from modifying data in applications with higher integrity levels.

Applications written specifically for Windows Vista will understand this nature of UAC fine; however, older applications running under Vista might fail to run, or might run in an unexpected manner. To allow for these older apps, Windows Vista offers virtualisation of key resources. The virtualisation essentially fools the older application into thinking that it has the required access to the necessary files/folder/Registry, but Vista is actually running those “simulated” resources in a safe and protected place.

Windows Vista includes file and Registry virtualisation technology for applications that are not UAC-compliant and that have historically required an administrator’s access token to run correctly. When such an application attempts to write to a protected directory, or to the Registry, Vista creates a virtual instance of the resource it is attempting to change. The virtualised copy is maintained under the user’s profile and is prevented from making actual changes to the system at large.